

Dayshaun Kakadiya

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 LinkedIn |  GitHub |  Codeforces |  LeetCode

Education

- **Sardar Vallabhbhai National Institute of Technology (NIT Surat)** Surat, Gujarat
B.Tech — Electronics & Communication Engineering | **CGPA: 9.08** 2023 – 2027
- **Gajera International School, Katargam, Surat** Surat, Gujarat
Class 12th - Percentage: **92.4%** | Class 10th - Percentage: **92.7%** 2021 – 2023

Projects

- **Multi-User Graph RAG Web App**   May 2026
Tools: Python, React, FastAPI, LangChain, NetworkX
 - Built and deployed a full-stack agentic AI app on Hugging Face Spaces for PDF-based Q&A with interactive knowledge graph visualization.
 - Designed a custom Graph RAG pipeline using SpaCy, Gemini embeddings, ChromaDB, and NetworkX, improving answer accuracy from 58% to 74%.
 - Implemented agentic graph traversal with LLM sufficiency checks, reducing retrieved context by 16% compared to vanilla top-k RAG while maintaining answer quality
 - Built a Dockerized multi-tenant backend with isolated vector stores and dynamic database provisioning, improving scalability and reducing response latency by 35%.
- **ByteLearn: Full Stack E-Learning Platform**   July 2025
Tools: Node.js, Express, MongoDB, JWT, Stripe, React.js, Tailwind CSS, Vercel
 - Implemented JWT based role-based authentication and authorization for instructors and students.
 - Developed RESTful APIs for course, user, and lesson management following the MVC architecture; optimized database queries to reduce API response time by 30%.
 - Designed scalable MongoDB schemas with Mongoose; delivered a fully responsive React/Tailwind UI, improving user engagement time by 20%.
 - Integrated Stripe for real-time course payments and deployed backend on Vercel for continuous delivery.
- **Multimodal Entity Extraction Engine**  December 2025
Tools: PyTorch, SmolVLM-256M, HuggingFace TRL, LoRA, DoRA, BitsAndBytes (4-bit), PEFT
 - Fine-tuned and self-hosted SmolVLM-256M-Instruct using QLoRA (4-bit NF4 quantization + DoRA) to extract structured product attributes from product images.
 - Applied PEFT with LoRA, training only 1.18% of parameters (3.07M/259.5M) using SFTTrainer.
 - Designed a custom `collate_fn` with label masking to compute loss only on assistant response tokens.

Skills

- **Programming Languages:** C/C++, Python, JavaScript
- **AI & Agentic Systems:** RAG pipelines, LangChain, LLM APIs (Gemini, HuggingFace), agentic reasoning loops, LoRA, QLoRA
- **Data Infrastructure:** ChromaDB, vector stores, spaCy, NetworkX, NumPy, pandas
- **Web Technologies:** React.js, Node.js, Express.js, FastAPI, MongoDB, REST APIs
- **Tools & DevOps:** Docker, Git, Linux, Vercel, Hugging Face Spaces, PyTorch, TensorFlow, HuggingFace
- **CS Fundamentals:** Data Structures & Algorithms, OOP, OS, DBMS

Achievements

- Author of a research paper accepted at **IEEE International Conference on Signal Processing and Communications (SPCOM 2026)**.
- Selected for **Amazon ML Summer School 2025**.
- **LeetCode Knight** (Max Rating: **1975**); solved **1000+** DSA problems across LeetCode, Codeforces, and CSES.